

Labwork 4: CPU vs GPU Grayscale Conversion

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Code

```
@cuda.jit
def grayscale_gpu_2d(src, dst) -> None:

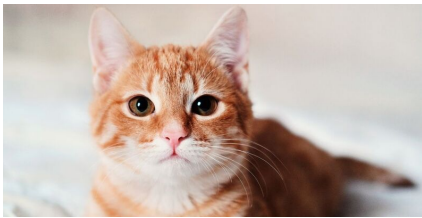
    tidx = cuda.threadIdx.x + cuda.blockIdx.x * cuda.blockDim.x
    tidy = cuda.threadIdx.y + cuda.blockIdx.y * cuda.blockDim.y
    g = np.uint8(
        (src[tidx, tidy, 0] + src[tidx, tidy, 1] + src[tidx, tidy, 2]) / 3)
    dst[tidx, tidy, 0] = dst[tidx, tidy, 1] = dst[tidx, tidy, 2] = g
```

Mean time by block sizes between 1D and 2D

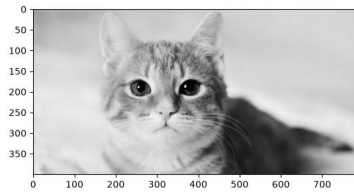
Table 1: Average CPU and GPU processing times by block size

Order	Dim	Threads	BlockSize	Time (ms)
0	1D	16	16	3.682668
1	1D	32	32	3.323858
2	1D	64	64	3.561551
3	1D	128	128	3.644618
4	1D	256	256	3.392350
5	1D	512	512	3.365658
6	1D	1024	1024	3.559546
7	2D	16	4:4	3.551440
8	2D	32	4:8	3.351309
9	2D	32	8:4	3.662337
10	2D	64	8:8	3.266118
11	2D	256	16:16	3.607598
12	2D	1024	32:32	4.407547

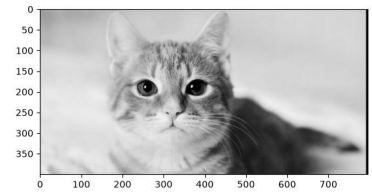
Original vs 1D vs 2D at block size 64



Original



1D GPU, 64



2D GPU, 64

Figure 1: Cat image comparison: original, 1D CUDA, and 2D CUDA results at block size 64.

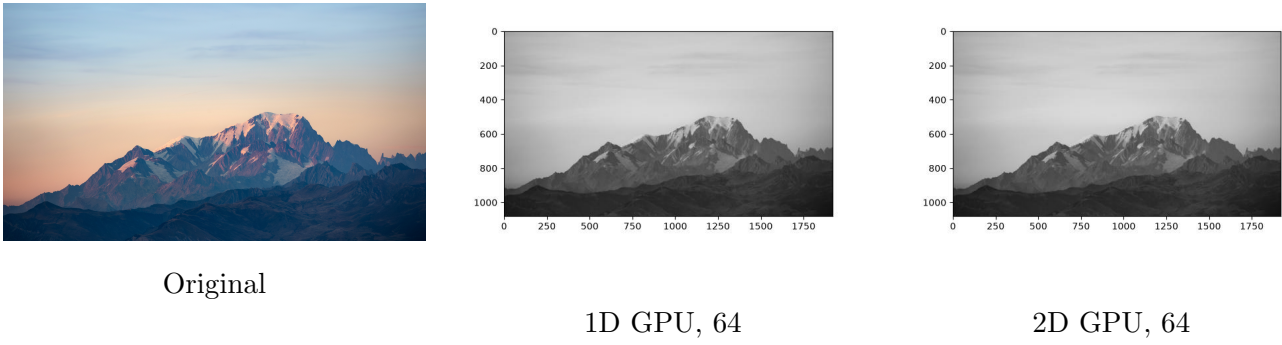


Figure 2: Wall image comparison: original, 1D CUDA, and 2D CUDA results at block size 64.

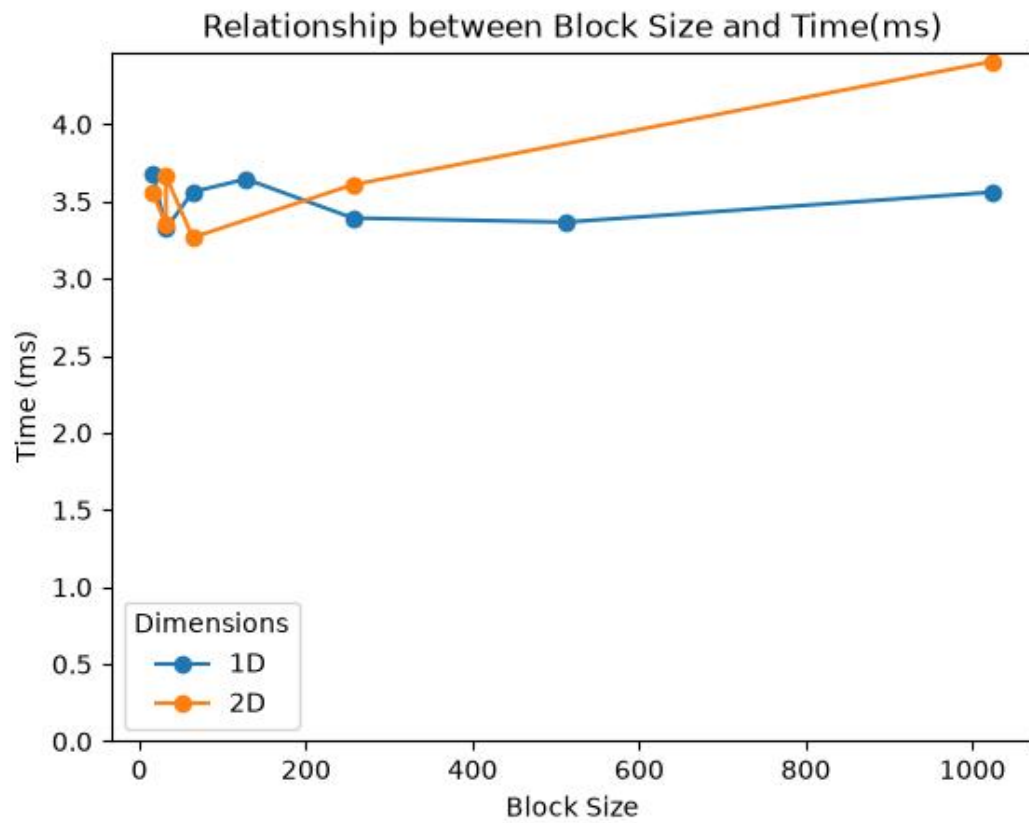


Figure 3: Mean time by block size and block dimension.